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PALEOMAGNETISM OF LOWER CARBONIFEROUS (TOURNAISIAN TO NAMURIAN) VOLCANIC ROCKS OF NEW BRUNSWICK, CANADA, AND ITS TECTONIC IMPLICATIONS

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ABSTRACT

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Following Kent and Opdyke's (1978) hypothesis, the possibility that a sinistral motion north of the Maine coastal volcanic belt has brought the New England–Canadian Maritimes region to its present position with respect to cratonic North America during the Carboniferous time was tested by a paleomagnetic study of gently dipping Lower Carboniferous volcanic rocks of New Brunswick. After removal of a low unblocking (T_{UB}) and low coercivity component variably present in the samples, a single coherent mean remanence direction was isolated: $D = 161°$, $I = +11°$, $K = 78$ and $\alpha_{95} = 5.5°$ for $N = 10$ sites (42 samples), characterized by a high T_{UB} (450°–660°C) linear vector trajectory present in each sample. The corresponding paleomagnetic pole falls at 136°E, 36°N ($d_m = 5.6°$, $d_p = 2.8°$). This pole is not significantly different from Early Carboniferous poles from the stable North American (NA) craton. If any left-lateral relative motion in the Carboniferous took place, it is not located in the Maritimes New England region but further north, closer to the craton, in the northern New Brunswick–southeastern Quebec region.

INTRODUCTION

The volcanic flows sampled for paleomagnetic purposes are located in two tectonic zones (Fig. 1): (1) on the New Brunswick Platform which forms a flat-lying Carboniferous cover unconformably overlying Silurian sedimentary rocks of the Fredericton Trough but overlapping the Cambro-Ordovician Miramichi Massif in the northeast; (2) within two Carboniferous outliers lying unconformably over Upper Silurian and Lower Devonian volcanic and sedimentary rocks on the southeast margin of the Late Ordovician to Early Devonian Matapedia Basin (Fyffe et al., 1981).

The Miramichi Massif lies within the Gander Zone of Williams (1978) which in New Brunswick is separated from the Precambrian Avalon Zone by the Fredericton

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Trough (Fig. 1). The northeastern Matapedia Basin contains a fault block of Ordovician ophiolitic rocks (Rast and Stringer, 1980) included within the Dunnage Zone (Williams, 1978). In the southwest, Upper Ordovician to Lower Silurian limestone of the Matapedia Basin overlies Cambro-Ordovician rocks similar to those exposed within the Miramichi Massif (Fyffe 1982).

The early Carboniferous volcanics and associated redbeds provide an opportunity to: (1) obtain a new reliable paleopole for Early Carboniferous time, (b) define more accurately the APW path of the northern Appalachians for this geological period; (c) test Kent and Opdyke's (1978, 1979, 1980) hypothesis that Acadia was situated some distance to the south of the present position relative to the North American (NA) craton in pre-Late Carboniferous time.

CARBONIFEROUS TECTONIC SETTING

The Carboniferous rocks of New Brunswick (Fig. 1) occupy a successor basin within the Appalachian Orogen (Gussow, 1953; Belt, 1968a, b; Webb, 1969; Hacquebard, 1972; Howie and Barss, 1975; Poole, 1976; Miall, 1978). The northwest margin of the successor basin is underlain by a thin, flat-lying sequence of Carboniferous continental sedimentary and minor volcanic rocks of the New Brunswick Platform (Belt, 1968b). To the southeast of the steep, northeast-trending Fredericton Fault, the gentle easterly slope of the Platform is interrupted by the Minto Uplift (Wade et al., 1977). The Marysville Subbasin lying to the west of this uplift contains evidence of a minor marine incursion (McCutcheon 1981).

The northeast-trending Belleisle Fault defines the boundary between the New Brunswick Platform and the deeper part of the Carboniferous basin to the south with its thick marine succession (Belt, 1968b). The Westmorland Uplift divides the marine basin into the Moncton and Cumberland subbasins. Rocks of the Moncton Subbasin have undergone at least two periods of pre-Pennsylvanian folding (Gussow, 1953).

Two outliers referred to as the Carlisle and Plaster Rock Basins lie to the northwest of the main outcrop area of Carboniferous rocks (St. Peter, 1979, 1982).

STRATIGRAPHY

Only the stratigraphy northwest of the Belleisle Fault is of concern here; several sections will be presented to illustrate the stratigraphic variation as it occurs across the area from the southeast to the northwest (Figs. 1 and 2).

The Carboniferous stratigraphy on the southern margin of the Marysville Subbasin has been established by MacKenzie (1964), Van de Poll (1967), Gemmell (1975), and Ruitenberg and McCutcheon (1980). The Carboniferous sequence, sitting unconformably on Silurian rocks of the Fredericton Trough, begins with a 25 m thick, discontinuous unit of felsic tuff that is underlain by a more widespread

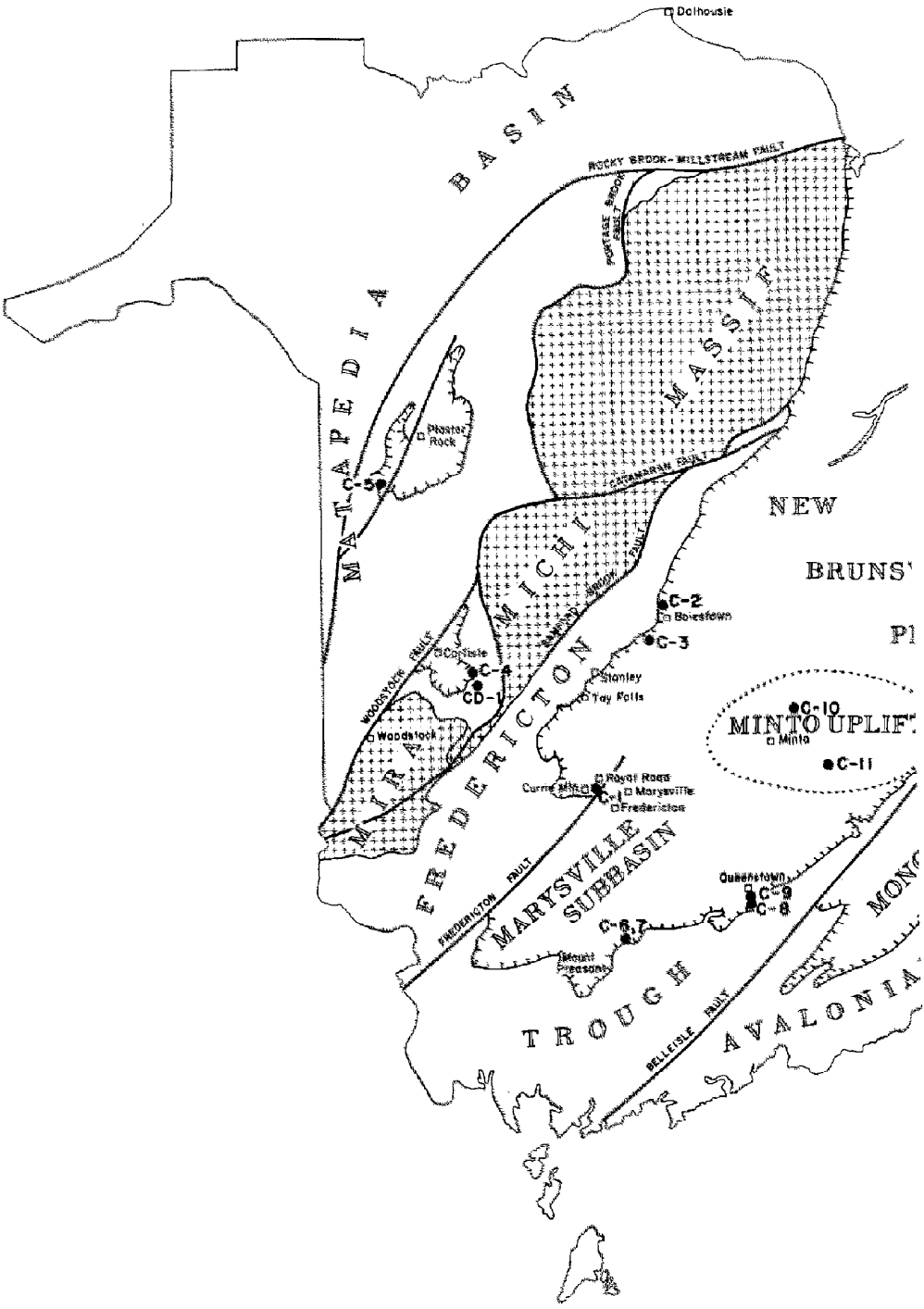
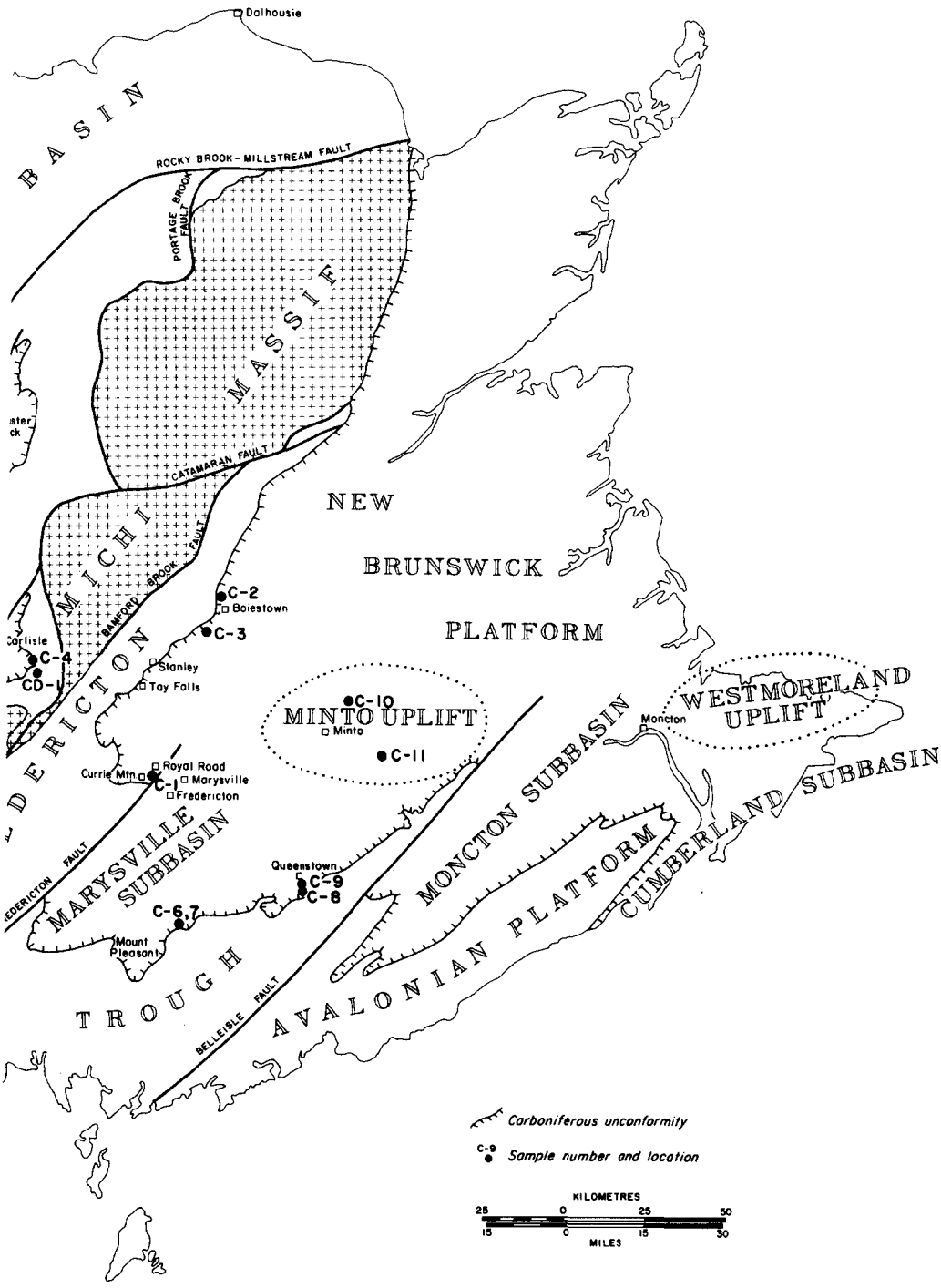


Fig. 1. Tectonostratigraphic zones of New Brunswick and distribution of paleomagnetic sampling sites.



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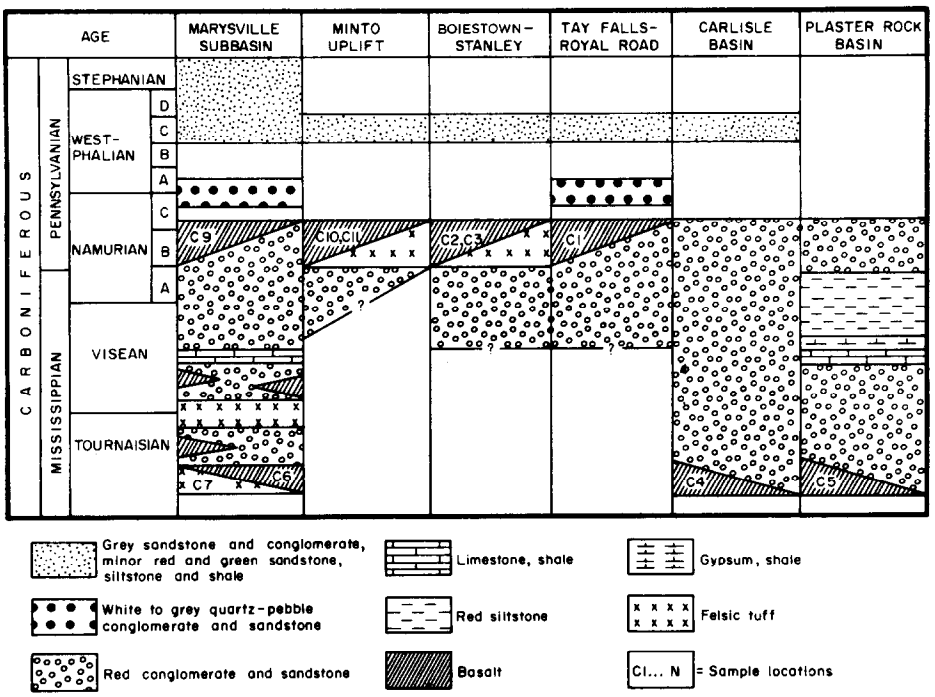


Fig. 2. Stratigraphic column of the Carboniferous period in New Brunswick and distribution of the sampling sites in the Tournaisian–Namurian interval.

basalt flow, approximately 50 m thick. According to Van de Poll (1967), the basalt consists mainly of plagioclase ranging in composition from labradorite to andesine. The matrix is mainly interstitial hematite and chlorite.

A unit of red conglomerate and sandstone up to 200 m thick overlies the basalt and contains two interbeds of basalt, each about 10 m thick. Approximately 200 m of felsic tuff succeeding these redbeds are in turn overlain by another 70 m unit of red conglomerate and sandstone which contains a single 15 m basalt flow. A 15 cm thick bed of argillaceous limestone of lower Visean (Windsor) age provides an important time marker at this point in the succession (McCutcheon, 1981).

Upper Namurian to lower Westphalian grey quartz-pebble conglomerate overlain by Westphalian-C grey sandstone and conglomerate are the youngest rocks preserved on the southern margin of the Marysville Subbasin.

A basalt flow interbedded with redbeds at a depth of 800 m in wells near Fredericton occurs 180 m above spores dated as Visean (M.S. Barss, pers. commun., 1982). This basalt is thus of similar age as the basalt at Queenstown on the southern margin of the basin.

The Carboniferous rocks on the basement high in the Minto area begin with a

basal red conglomerate which becomes finer upward and is capped by a thin unit of greenish-grey felsic tuff. Grey, fine-grained basalt, approximately 30 m-thick, is exposed above the tuff. The basalt consists principally of plagioclase; magnetite is abundant, and clinopyroxene, olivine and hematite are present in small amounts (Dyer, 1926; Muller, 1951).

Grey, green and red sandstone, conglomerate and shale with a preserved thickness of 250 m overlies the basalt. The lower part of this post-basalt sequence is dated as Westphalian-C (Muller 1951).

Riebeckite–aegirine-bearing rhyolite and basalt exposed on hills at the east end of Grand Lake have been interpreted by Muller to overlie the Pennsylvanian rocks but are more likely older (Van de Poll, pers. commun., 1981).

The main belt of Carboniferous rocks on the New Brunswick Platform to the north of the Fredericton Fault lies unconformably upon Silurian rocks of the Fredericton Trough. The Silurian rocks are in fault contact with Cambro-Ordovician rocks of the Miramichi Massif to the northwest along the northeast-trending Bamford Brook Fault (Irrinki, 1980).

A sequence of red conglomerate and sandstone up to 400 m thick forms the basal unit of the northwestern margin of the Platform. The redbeds range in age from Visean to Namurian (Van de Poll, 1973). At least two basalt flows, each about 15 m thick, are interbedded with the upper part of the redbed sequence near Boiestown in central New Brunswick. The basalts pinch out to the southwest where a tuff is interbedded with grey sandstone, conglomerate and siltstone sitting on the redbeds. Grey sandstone and conglomerate of Pennsylvanian age overlie the basalt and tuff (Poole, 1958; Ball et al., 1981).

Farther south, northwest of Fredericton, volcanics are absent and a quartz-pebble conglomerate, overlies and locally oversteps the redbeds. Grey sandstone of Westphalian-C age lies above the conglomerate (Poole, 1957).

Just northwest of the Fredericton Fault at the Royal Road quarry, two 15-m thick basalt flows separated by 20 m of red conglomerate and sandstone are overlain by a conglomerate of upper Namurian–lower Westphalian-A age (Van de Poll, 1973). These basalts exposed on the surface at the Royal Road quarry occur at the same stratigraphic position as the previously described basalt found at a depth of 800 m, south of the Fredericton Fault.

The Carboniferous rocks of the Plaster Rock area unconformably overlie Siluro-Devonian rocks of the Chaleur Bay Synclinorium (Ruitenberg et al., 1977), which forms the eastern margin of the Matapedia Basin. The basal unit of the Plaster Rock Basin is a columnar 30-m thick basalt. The basalt is porphyritic and amygdaloidal. Phenocrysts of labradorite and clinopyroxene are set in a matrix of plagioclase, calcite and minor opaque minerals.

The basalt is overlain by a 550-m thick alluvial fan and fluviatile sequence of red conglomerate and sandstone that is succeeded by 10 m of caliche-type nodular limestone and shale and 30 m of diagenetic nodular gypsum and shale. Red siltstone

and conglomerate, about 275 m thick, are the youngest rocks preserved in the basin (Rose, 1936; St. Peter, 1979). The paleoclimatic conditions recorded for the Plaster Rock Basin parallel those existing during the Mississippian in southern New Brunswick (St. Peter, 1979).

No fossils have been recovered from the Plaster Rock Basin, but lithological similarities suggest correlation of the lower redbeds with the Tournaisian Horton Group of southern New Brunswick, the limestone and gypsum beds with the Visean Windsor Group and the upper redbeds with the upper Visean to Namurian Hopewell Group (Rose, 1936). If the above correlations are correct, the basalt in this area is Tournaisian.

The Carlisle Basin is situated on a graben that effectively divides the Cambro-Ordovician rocks of the Miramichi Massif into two blocks. Flat-lying Carboniferous redbeds on the southern and eastern margin of the Carlisle Basin rest unconformably on Devonian volcanic rocks, whereas redbeds on the western margin dip up to 20° westward and are separated by the Woodstock Fault (Venugopal, 1981) from Upper Ordovician, Silurian, and Devonian sedimentary rocks of the Matapedia Basin.

Basalt, about 20 m thick, deposited nonconformably on Devonian gabbro comprises the lowermost Carboniferous unit within the Carlisle Basin (Anderson, 1968; St. Peter, 1982). The basalt contains large vesicles and displays a subhorizontal layering defined by the alignment of small plagioclase laths. No ferromagnesian minerals are preserved. Scattered irregular grains of hematite forming up to 15% of the rock are interstitial to plagioclase. Vesicles are filled with calcite and chlorite.

Red conglomerate and sandstone overlying the basalt form alluvial fan deposits between 150 and 200 m thick. Grey fluvial sandstone and conglomerate with a preserved thickness of approximately 100 m gradationally overlie the redbeds. St. Peter (1982) reports *Calamites* sp. in the grey beds which indicates a Pennsylvanian age for them. The underlying redbeds and basal basalt can be correlated with the similar Tournaisian to Namurian rocks of the Plaster Rock Basin. Venugopal (1982) notes the presence of thin beds of nodular limestone within the redbeds on the southern margin of the Carlisle Basin which may correlate with the caliche beds of the Plaster Rock Basin.

PALEOMAGNETIC SAMPLING AND MEASUREMENTS

In the present paleomagnetic study, 62 hand oriented samples at 12 sites (with a minimum of five samples per site) were collected from mainly basaltic volcanics emplaced within paleontologically well-dated sediments in central and southern New Brunswick (Fig. 2).

The magnetization was measured with a digital Schonstedt spinner magnetometer (DSM-1) coupled to PDP-1 and APPLE II minicomputers in tandem; the volume susceptibility was measured at room temperature in the Earth's field ($H = 0.05$ mT),

using a Cenco MS-3B susceptibility bridge. Two methods of demagnetization were carried out using (1) an AF demagnetizer surrounded by three large concentric mu-metal cylinders with thee-axis tumbler operating in residual fields of about 5 mT (Seguin, 1975) and a Schonstedt GSD-1 apparatus (100 mT peak-to-peak intensity) and (2) a Schonstedt thermal demagnetizer (TSD-1) with residual fields of less than 15 mT. Stepwise AF and thermal procedures were employed up to 100 mT and 660°C respectively. For the NRM of each oriented sample, at least two specimens (diameter 2.5 cm, height 2.2 cm) were used to obtain a sample mean direction. Single measurements of NRM were usually repeatable to within 4% in intensity (J) and within 3° in direction (declination) D, inclination I); the total estimated uncertainty in the NRM determinations (combined orientation and measurement error) was about 5% in intensity and 3° in direction.

NRM OF VOLCANIC AND DYKE ROCKS

Individual specimen NRM directions of Carboniferous volcanics and associated dykes from New Brunswick, although characterized by wide scatter of positively and occasional negatively inclined steep and shallow remanences, have a pronounced deviation away from the present Earth's field (PEF) ($D = 338°$; $I = 74°$) at the site localities indicating that these rocks carry some ancient remanence particularly since the directions of the remanence of the few specimens ($\sim 10\%$) that lie close to PEF were found to move away from those initial directions during subsequent AF or thermal treatment (Fig. 3 and Table 1). The intensities of NRM (J_n) vary from 4×10^{-3} to 8 Am^{-1} with the values of Koenigsberger ratio ($Q_n = J_n/KH$) between 0.2 and 5 but with predominant ratios between 0.4 and 2 (Fig. 3).

PRELIMINARY DEMAGNETIZATION STUDIES

For the preliminary AF study, at least five (5) samples per site were demagnetized in 10 to 12 steps up to 100 mT, and vector diagrams (Zijderveld, 1967) were used in the subsequent analysis. For 75% of the samples, the intensities of magnetization fall off to less than 20% of the original NRM intensity in the interval 25–100 mT (average = 70 mT). The median destructive field ($H_{1/2}$) varies from 10 to 100 mT (average = 30 mT). In almost 50% of the samples (25/59), the measurements indicated acquisition of spurious magnetizations at higher demagnetizing fields (60–100 mT; average = 85 mT). Generally, vector trajectories are linear until stable end-points are reached at higher fields and an univectorial component (E, shallow positive) is obtained once the low coercivity (soft) component is removed in the 10–40 mT range (average = 25 mT). Specimen C9-5A is a representative example of such behaviour (Fig. 4A). Most commonly, the AF demagnetization is incomplete and insufficient. Some samples (see Table 1) are characterized by mixed and reversed polarities and one site (C-3) is characterized by a predominantly normal polarity upon AF demagnetization.

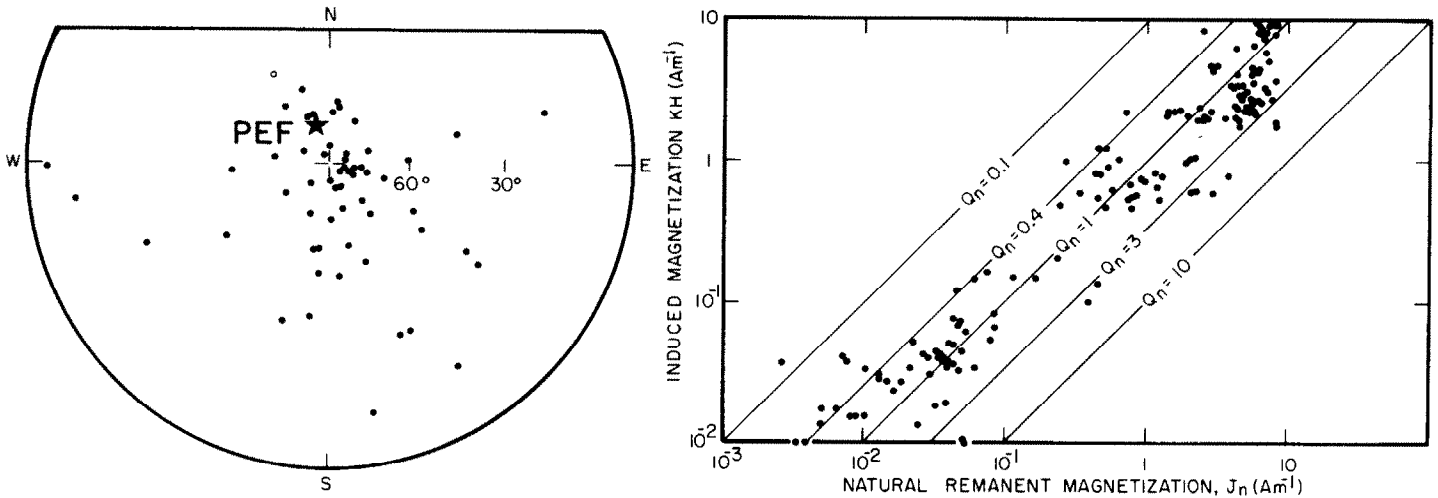


Fig. 3. Sample NRM directions plotted on Wulff net for the New Brunswick Carboniferous volcanic rocks (left). Solid (open) circles represent projections (lower (upper) hemisphere). The star represents the present Earth's field (PEF) at the sampling region (338°, 72°N). Plots of intensity of NRM (J_n in A/m) for individual specimens against the induced magnetization (KH in A/m) with resulting slope lines indicating the values of Koenigsberger ratio, Q_n (right).

TABLE 1

Site-mean directions of magnetization *

Site	Formation		N_1	n	In-situ				Tilt-corrected		Locality	
	strike^0	dip^0			D^0	I^0	α_{95}^0	k	D^0	I^0	(lat. °N	long. ^{0}W)
Natural Remanent Magnetization												
C-1	359	15E	6	16	280	83	33.1	1.9	080	82	46.00	66.69
C-2	010	04E	5	17	248	73	30.8	7.2	239	76	46.48	66.47
C-3	022	23E	5	13	346	88	26.2	9.5	108	68	46.40	66.48
C-4	000	00	5	9	074	74	17.4	20.4	074	74	46.27	67.20
C-5	059	04NW	5	16	085	53	49.8	3.5	080	55	46.77	67.56
C-6	020	12W	5	14	150	54	21.6	13.5	164	62	45.58	66.57
C-7	069	20NW	4	8	157	75	18.1	26.6	346	85	45.58	66.57
C-8	Basaltic dyke		5	13	156	26	23.8	11.4	156	26	45.67	66.10
C-9	359	15E	5	12	248	72	40.5	3.4	195	83	45.68	66.10
C-10	000	00	5	13	121	83	9.1	72.5	121	83	46.18	66.03
C-11	000	00	5	14	195	62	17.9	19.3	195	62	46.03	65.87
CD-1	dyke		5	15	306	78	35.0	4.0	306	78	46.26	67.25

$N = 12$

in-situ: 159,79; $\alpha_{95} = 14.2$; $R = 10.95$; $K = 10.4$

tilt-corrected: 136,79; $\alpha_{95} = 13.5$; $R = 11.03$; $K = 11.4$

Mainly thermally and/or AF demagnetized

	N_2	n	T_{UB} range	In-situ				Tilt-corrected		Polarity
			(°C)	$D_1°$	$I_1°$	$\alpha_{95}°$	k	D_2	T_2	
C-1	5	11	575–630	164	12	13.5	33	163	11	R
C-2	5	14	500–630	157	22	12.1	40	155	20	R
C-3	4	8	450–600	160	10	11.8	61	155	−05	M (N)
C-4				cannot be cleaned adequately						
C-5	2	6	600–630	166	08	–	–	166	05	M
C-6	5	13	500–630	163	06	14.2	30.4	162	13	R
C-7	3	5	500–535	157	12	20.1	37	154	30	R
C-8	5	13	525–630	160	11	18.6	17.9	160	11	R
C-9	3	7	450–600	153	07	22.6	30.8	153	06	M (R)
C-10	5	12	570–630	158	21	7.1	115	158	21	R
C-11	5	13	500–660	174	−02	8.6	79	174	−02	R
CD-1	42	102		unstable						

	N	$D°$	$I°$	R	k	$\alpha_{95}°$
Thermal, in-situ:	10	161	11	9.885	78.3	5.5
Thermal, tilt-corrected:	10	160	11	9.801	48	7.3

* N_1, and N_2 are the number of samples used in the statistics at NRM and after thermal treatment respectively; n is the number of specimens used in each case; $D°$, $I°$ are declination and inclination of remanent magnetization; k is the estimate of Fisher's precision parameter; $\alpha_{95}°$ is semi-angle of cone of 95% confidence for mean direction: R is the resultant of N unit vectors. T_{UB} range = range of unblocking temperatures over which the directions are computed from linear trajectories in Zijderveld vector diagrams. R = Reverse polarity; M = Mixed polarity; M (R, N) = Mixed polarity (predominantly reverse or normal).

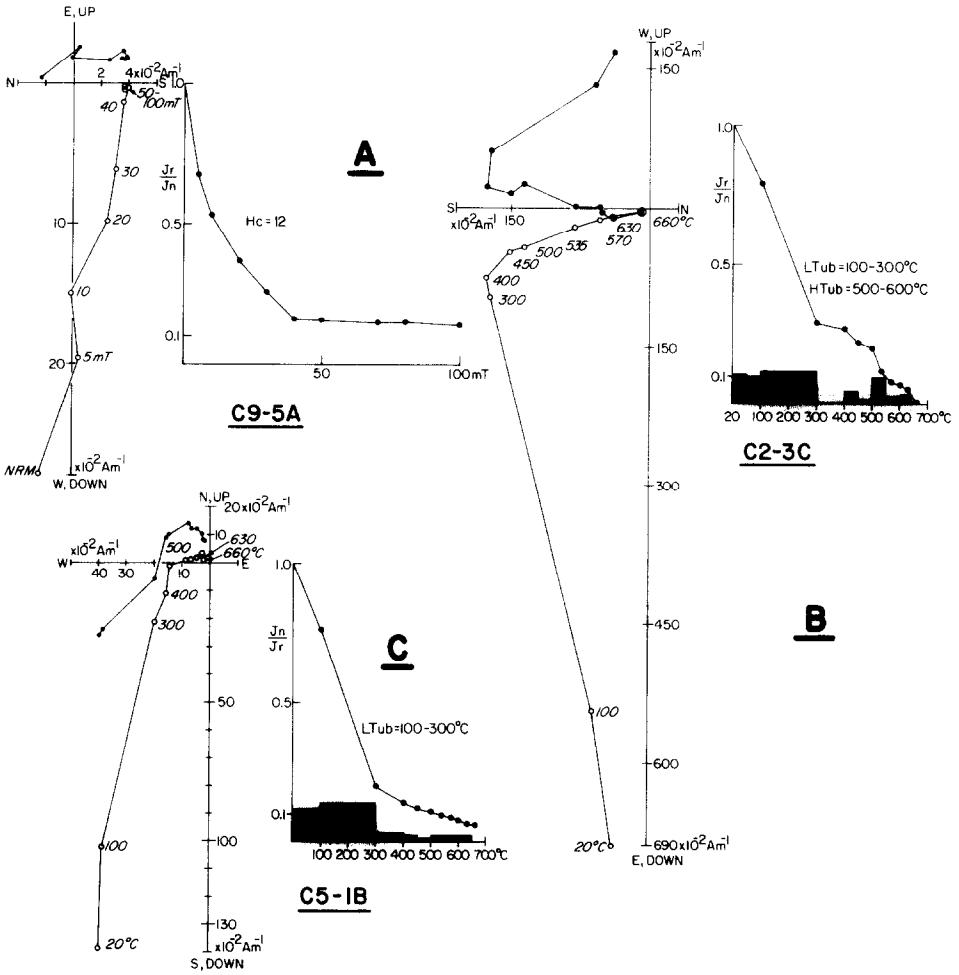


Fig. 4. Representative orthogonal AF (a) and thermal (b) demagnetization diagrams (Zijderveld, 1967) of successive points of magnetization for three basaltic specimens as indicated. Solid (open) circles represent projections on horizontal (vertical) plane. Numbers next to open circles represent the demagnetization steps (in mT or °C).

Petrographic examination by transmitted and reflected light was used in order to elucidate the nature of the magnetic memory carrier. Most of the sites studied consist of fresh volcanic flows; sites: C-4, C-7, C-9 and to a lesser extent C-5 are exceptions to this general rule. Sites CD-1 and C-4 are considerably altered, whereas sites C-7 and C-9 are moderately altered. The pyroxenes and amphiboles are chloritized, the feldspars sericitized and part of the magnetite and hematite transformed to limonite. Site C-5 is only slightly altered but contains numerous vesicles partly filled by secondary material.

Preliminary thermal demagnetizations of some specimens from the samples used

in AF treatment and approximately the remaining 60 percent of the whole collection were made in order to obtain more information about the component of magnetization. Ten to twelve specimens for each of the 12 sites were thermally cleaned in a maximum of 11 steps up to 660°C and vector diagrams (Zijderveld, 1967) were used in the subsequent analysis of the results. Most commonly, the blocking temperature (T_{UB}) is located in the 350°–610°C range (mean = 530°C). Approximately 35% of the T_{UB}s are higher than 600°C (600°–660°C). A characteristic univectorial component (SSE, shallow positive) is obtained as depicted by linear vector trajectories after removal of a soft component of recent origin in the 100°–400°C range. Typical behaviour of a specimen (C2-3C) containing this component in the course of thermal treatment is shown in Fig. 4B. In more than 80% of the cases, the AF demagnetized and thermally cleaned component on the samples coincided. Samples from sites C-1, C-2, C-6, C-7, C-10, and C-11 are characterized by a single reverse thermally cleaned component. Mixed polarities are observed on sites C-3, C-5, and C-9. Normal polarities are isolated in samples 3, 4 and 5 of site C-3, sample 1 of site C-5 (specimen C5-1B, Fig. 4C) and sample 3 of site C-9. Mixed polarities are observed in specimens of sample C-3-1.

In summary, the most important feature of the Carboniferous volcanics of central and southern New Brunswick is the presence of one dominant remanent magnetization oriented SSE.

STATISTICAL ANALYSIS OF SITE-MEAN DIRECTIONS

Site-mean directions for the various sites are shown in Table 1. For the 12 sites considered, the mean direction is 159,79 ($\alpha_{95} = 14.2°$, $k = 10.4$) for the formations in-situ and 136,79 ($\alpha_{95} = 13.5°$, $k = 11.4$) after tilt correction. The site-mean directions obtained after AF and thermal treatment are listed in Table 1 for both in-situ and tilt-corrected formations and are shown in Fig. 5 for tilt-corrected formations (when needed). The mean direction for the cleaned component on 10 sites (excluding C-4 and CD-1) is 161,11 ($\alpha_{95} = 5.5°$, $k = 78$) for the formations in-situ and 160,11 ($\alpha_{95} = 7.3°$, $k = 48$) after tilt correction.

All sites are basaltic lava flows except site CD-1 which consists of monzonite, diabase and trap dykes, site C-7 represented by a relatively altered felsic tuff and site C-8 which is a basaltic dyke intruding lower Carboniferous felsic volcanics. For the cleaned component, sites C-4 and CD-1 were removed from the statistics; C-4 could not be cleaned satisfactorily and CD-1 turned out to be unstable.

Appreciable variations in declination and inclination as well as polarity changes in the volcanic rocks can either indicate differences of primary ages or else subsequent overprint. As both normal and reverse polarities are observed in this study, this suggests that the data of acquisition of magnetization is older than the Westphalian-B magnetic marker horizon of Roy (1977). This indicates that the magnetization is likely primary. The tilt correction which is negative (but not at a

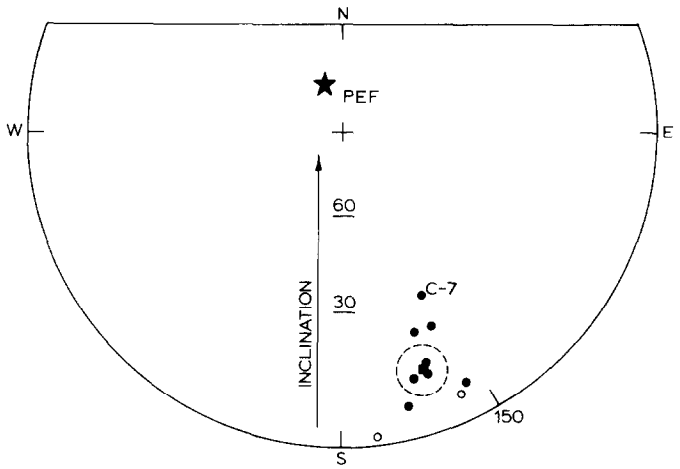


Fig. 5. Site-mean directions of magnetization plotted on Wulff net for volcanic rocks (Table 1) after AF and thermal treatment. The representative tilted site-mean directions with positive inclinations are represented by full circles and those with negative inclinations by open circles. The mean value of the cleaned component is indicated by a square. The large dashed circle represents the angle of cone of 95% confidence for the mean direction. Site C-7, an acidic tuff, is labelled.

significant level) is in apparent contradiction with the above statement.

The crucial importance of a negative fold test that is not significant warrants a quantitative approach. Using the statistical method of the fold test suggested by McElhinny (1964) with $N = 10$, $k_2/k_1 = 78.3/48 = 1.6312$ (Table 1), it is clear that the fold test is not significantly negative at the 95% level of confidence. Recently, McFadden and Jones (1981) have demonstrated that the test indicates significance under far less severe folding than does the test suggested by McElhinny. It is thus more appropriate to make use of this test. To do so, it was decided to test whether the overall mean directions from the different limbs may be distinguished statistically. In this last test, $N = 7$ with four sites (C-1, 2, 3, 9) dipping to the east (E) and three sites (C-5, 6, 7) dipping to the west (W)-northwest. The resultant $R_E = 3.9456$ and the $R_W = 2.9406$. For the in-situ directions, the resultant $R_1 = 6.9547$ and for the tilt-corrected directions, $R_2 = 6.8658$. Using $R = R_2$, the value of 0.206 is smaller than 0.821 from Table 1 of McFadden and Jones (1981) and so the hypothesis of a common true mean direction after unfolding cannot be rejected at the 95% confidence limit. With $R = R_1$, the value of 0.605 is also smaller than 0.821 but very close to it. There also, the hypothesis of a common true mean direction in-situ cannot be rejected at the 95% confidence limit. Hence the negative fold test is a possibility but remains statistically unproven.

However, there are definite indications that the flow layering was not originally horizontal but made an angle of 5° to 10° with respect to the horizontal in the vicinity of volcanic centers (Muller, 1951). In addition, the dip angles are generally

so small that the reliability of a fold test is questionable. The question is what is the true age of magnetization of the New Brunswick Carboniferous units investigated in this study? Primary or secondary? What are the criteria to decipher a primary component or an overprint? One can try to categorize the sites of Carboniferous units on the basis of different geological setting or age to detect any trend in magnetization. Other criteria include (a) a degree of parallelism between reverse and mixed polarities, (b) angular dispersion of individual sites within a group before and after tilting, (c) angular dispersion between groups of different ages, etc...

The angular dispersion of the cleaned component is not very large, but it may be increased by the presence of mixed polarities in these volcanic rocks. In order to test this possibility, a separate statistics was carried out for sites having a reverse polarity and those having mixed polarities. The first category includes sites C-1, C-2, C-6, C-7, C-8, C-10 and C-11 while the second includes sites C-3, C-5 and C-9. The mean directions are 160,08 ($\alpha_{95} = 10.1°$, $k = 150$) and 162,12 ($\alpha_{95} = 7.6°$, $k = 63$) for the two categories (M and R) respectively. As expected, the dispersion is smaller for the reverse than mixed polarity groups. Given the relatively large α_{95} of both categories, no significant difference in direction of remanence is observed and this almost perfect antiparallelism is suggestive of a primary origin of magnetization. The angular difference (in degrees of arc) between the direction of each individual site and the mean direction before and after tilt correction is about the same except for site C-7 where it is much larger after tilt (23° vs 5°). This analysis indicates that the magnetization of this site (which is a tuff and not a flow) is secondary.

One can then try to categorize the sites of Carboniferous basalts on the basis of different geological setting or age to detect any change in trend of magnetization.

The Miramichi Massif (MM) is the most striking geological feature separating the northern and southern Carboniferous sites in New Brunswick. The sites north of the MM are C-4 and C-5 while those south of it are C-1, C-2, C-3, C-6, C-7, C-8, C-9, C-10, and C-11. Sites C-4 and C-7 are excluded for reasons mentioned above. The tilt corrected mean direction of the cleaned component to the north of MM is 166,08 and 166,05 (before and after tilt correction) and to the south 162,11 and 159,12 (before and after tilt). It is noted that the direction of magnetization from the site north of MM is similar to that of the eight sites south of MM; site C-9 is possibly the only exception; it is located just north of a fault (McCutcheon, 1981). This similarity in direction of magnetization suggests the absence of a rotation around a sub-vertical axis of the block northwest of MM with respect to MM or else strike-slip displacement along faults bounding the MM. A possible candidate for the sinistral movement proposed by Kent and Opdyke (1978) is the Fredericton fault. Sites C-1, C-2, C-3 and C-5 are located north of the Fredericton fault while sites C-6 to C-11 are located south of it. The mean tilt-corrected direction of magnetization for the northern and southern groups is 160,08 ($\alpha_{95} = 13°$, $k = 45$) and 160,13 ($\alpha_{95} = 11°$, $k = 37$). As the hypothetical left-lateral movement ought to be located north of the Bellelisle Fault, it can now be added that it also ought to be north of the

Fredericton Faults and the Miramichi Massif, that it as far north as the Matapedia Basin. On geological ground, the faults observed in the Fredericton Trough suggest a right-lateral rather than left-lateral motion.

Sites C-5, C-6, and C-7 are of Tournaisian age (lower Mississippian) while sites C-1 to C-3 and C-9 to C-11 are of Namurian age (lower Pennsylvanian). Site C-5 located on the border of the Matapedia Basin is not significantly different from sites C-6 and C-7 and has apparently not been rotated locally. Based on the available data, it is not possible to detect a polar drift between the Tournaisian [162,09 ($\alpha_{95} = 8.3°$, $k = 221$) in-situ and 162,16 ($\alpha_{95} = 21°$, $k = 35$) tilt-corrected] and Namurian [161,12 ($\alpha_{95} = 9.6°$, $k = 50$) in-situ and 161,09 ($\alpha_{95} = 11.1°$, $k = 39$) tilt-corrected]. The two closest sites, C-6 (Tournaisian; 163,06) and C-9 (Namurian; 153,07) suggest a certain amount of drift but site C-9 is near a fault and a small amount of local rotation is possible.

DISCUSSION

A compilation of Late Carboniferous/Early Permian and Early Carboniferous paleomagnetic poles for North America allows a comparison between the stable interior of North America and the Maritime Appalachians (Table 2). The mean paleopole position from the Early Carboniferous New Brunswick basalts (137°E, 36°N, $d_m = 5.6°$, $d_p = 2.8°$) is considered as a reliable pole considering the relatively small amount of dispersion involved and is similar to that of the stable NA craton (131°E, 41°N) and to the Late Carboniferous poles of the Maritime Provinces (Fig. 6). The paleolatitude is 6°N.

Kent and Opdyke (1978, 1979, 1980) have suggested that the Siluro-Devonian volcanics and unconformably overlying Devonian Perry Formation of coastal Maine and New Brunswick are part of an Acadian displaced terrain that includes the Avalonian Platform to the southeast. They further surmise that a zone of Carboniferous transcurrent motion must be located northwestward of this volcanic belt. However, our results show that during the Early Carboniferous, this area (the southern margin of the Fredericton Trough) was just as likely part of North America. This is supported by the fact that many of the faults north of the volcanic belt are intruded by early Devonian plutons and hence, could not have undergone large transcurrent displacements during the Late Devonian or Early Carboniferous (Fyffe et al., 1981; Ludman, 1981). In addition, most of the faults (e.g. Fredericton fault) are characterized by a right-lateral component of motion and not a left-lateral component; the largest fraction of the motion is vertical.

Irving (1979) has summarized the evidence for displaced terrains in North America and notes the lack of data prior to the late Carboniferous. Roy and Morris (1983) argue that the limited Carboniferous data set for the east coast Carboniferous basin and NA craton is not precise enough in terms of the contemporaneity of ages of magnetization of the respective poles to detect the extent of any motions of

TABLE 2

Paleomagnetic Pole Positions

Symbol	Rock unit		Pole position (°N)	Pole position (°E)	Reference
North American cratonic Sites					
Late Carboniferous (Stephanian) to Early Permian					
IN	Ingelside Fm.		46	122	Diehl and Shive (1979)
CA	Casper Fm.		47	127	Diehl and Shieve (1981)
BC	Brush Creek		35	125	Payne et al. (1978)
CT	Cutler Fm.		44	119	Helsley (1971)
EC	Elephant Canyon		44	120	13.52 *
HA	Halgaito Tongue		49	117	11.67 *
MR	Maroon Fm.		41	124	Christensen (1974)
DU	Dunkard Fm.		44	122	Helsley (1965)
Average ($N = 8$):		A1	44	121	
Late Carboniferous (Westphalian)					
CP	Coal Pole	A2	37	132	Kopacz and Noltimier (1976)
Early Carboniferous (Tournaisian and Visean)					
MC	Mauch Chunk Fm.		43	127	Knowles and Opdyke (1968)
SJ	St. Joe Lms.		39	132	Scott (1979)
Average ($N = 2$)		A3	41	130	
Canadian Maritime Provinces					
Late Carboniferous (Stephanian)					
PE_1	Prince Edward Island redbeds		40	127	8.95 *
PE_2	Prince Edward Island redbeds		42	133	8.99 *
Average ($N = 2$)		B1	41	130	
Late Carboniferous (Westphalian)					
BO	Bonaventure Fm.		38	133	8.100 *
HC	Hurley Creek Fm.		39	125	9.998 *
CU	Cumberland Gp.		36	125	11.76 *
PI	Pictou Gp ss.		41	132	8.101 *
Average ($N = 4$)		B2	39	129	
Early Carboniferous (Tournaisian to Namurian)					
HW	Hopewell Gp.		34	118	Roy and Park (1969)
MP	Minudie Point		35	124	Roy (1977)
MF	Maringouin Fm,		34	117	Roy and Robertson (1968)
Average ($N = 3$)		B3	34	120	
BV	New Brunswick volcs,		36	136	this study

* Numbered references are from McElhinny's list in the Geophysical Journal 1968–1978, "Paleomagnetic Directions and Pole Positions".

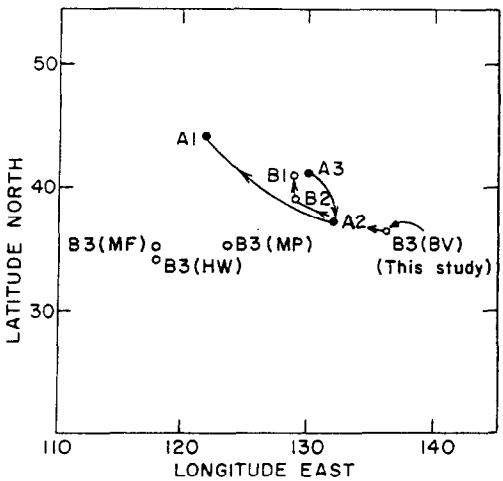


Fig. 6. Paleomagnetic Carboniferous poles for: (a) cratonic North America, (b) east coast of North America (Table 2).

Acadia relative to the craton during a short time interval. Kent and Opdyke (1980) and Roy (1982) provide evidences for and against the relative displacement of Acadia and the NA craton in Carboniferous time. Given that both reverse and normal polarities are observed in the Early Carboniferous, it is more difficult to obtain an accurate mean direction of magnetization as the pole and antipole need not be perfectly antiparallel. Adding these two inaccuracies to the one related to secular variations yields paleomagnetic results lacking sufficient resolution to prove the presence of longitudinal displacements as postulated by Kent and Opdyke. Indeed, Robertson et al. (1968) suggest an error of 15° in the position of the Late Devonian Perry pole—one of the poles used by Kent and Opdyke (1978) to calculate some 1500 km of left-lateral movement along northeast-trending faults to produce a displacement of 15° of latitude.

It is not our intention to deny the existence of displaced terrains in the Appalachians at Late Carboniferous time. We maintain only that our results are consistent with the Maine volcanic belt already being docked to eastern North America in the early Carboniferous in accordance with the geologic constraints. Our conclusions are in conflict with several previous interpretations and indicate that Appalachian paleomagnetic studies are neither plentiful nor accurate enough at present to outline displaced terrains in space and time during the Carboniferous.

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